

ERP System — Business Flow

Internal Documentation — Confidential

Version	Date	Author	Changes
1.0	June 2026	System	Initial release
1.1	June 2026	System	Gap fixes: putaway assignment, checker permissions, failed delivery resolution, attendance late detection, push notifications

Modules: AUTH · HRIS · INVENTORY · WMS · TMS · FINANCE

Overview

The ERP system is composed of six modules. WMS is the operational hub — it sits at the intersection of inventory, logistics, and finance, orchestrating the core fulfillment pipeline.

Module	Responsibility
AUTH	JWT-based login, token refresh, and user profile. Foundation for all other modules.
HRIS	Employee records, shift scheduling, attendance tracking (present/late/absent), and login audit logs.
INVENTORY	Product catalog, warehouse bin locations, and real-time stock levels. Central ledger for all physical goods.
WMS	Warehouse operations — inbound receiving & putaway, outbound picking, QC checking, dispatch, and invoice generation.
TMS	Transportation — route planning, GPS tracking, delivery confirmation (POD), failed delivery resolution, and driver fuel logging.
FINANCE	Accounts Payable (AP) and Accounts Receivable (AR) — invoices, payments, aging summaries, and branch management.

Cross-Module Dependencies

Dependency	Description
WMS → INVENTORY	Receiving goods adds stock; confirming picks deducts stock from bins.
WMS → TMS	Dispatching a sales order triggers creation of a TMS delivery route.

WMS → FINANCE	Fulfilled SOs generate AR invoices; POs feed AP invoices.
HRIS → WMS / TMS	Employee records used when assigning pick tasks, putaway, check tasks, and driver routes.
TMS → FINANCE	Cancelled failed deliveries void the associated AR invoice.

Flow 1 — Procurement-to-Pay

Covers the inbound goods journey from supplier purchase order through to payment. Managed by Operations Managers, Warehouse teams, and Finance.

Inbound Receiving Workflow — WMS + INVENTORY + FINANCE

Step 1 — Create Purchase Order *(WMS)*

Operations Manager creates a PO with supplier and product lines. Status: `pending`.

Step 2 — Receive Goods at Warehouse *(WMS)*

Warehouse Supervisor or Operator records received quantities against the PO. A Goods Receiving Note (GRN) is created. Stock is added to INVENTORY at a staging bin.

Step 3 — Generate Putaway Tasks *(WMS)*

Supervisor selects one or more GRN lines and assigns them to a specific Warehouse Operator. The system creates putaway tasks with status `pending`, linked to the GRN, the operator, and the source warehouse. An operator can only see tasks explicitly assigned to them.

****v1.1 clarification:**** Putaway tasks are supervisor-initiated and operator-specific. The operator does not self-assign — they receive tasks in their mobile queue. The system validates that the assignee has the ``wh_operator`` role. Duplicate tasks for the same GRN line are prevented.

Step 4 — Execute Putaway *(WMS · INVENTORY)*

Operator opens the Putaway screen on mobile. Only their assigned tasks are shown by default. For each task, the operator selects or scans a target bin. On confirmation:

- Stock is moved from the staging bin to the target bin in a single transaction.
- Task status becomes `completed`.
- INVENTORY bin levels are updated.

Step 5 — Create & Pay Supplier Invoice *(FINANCE)*

Finance Officer creates an AP invoice linked to the PO. Payment is recorded when processed, updating the AP aging report.

Roles Involved

Step	Role	Action
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Create PO	operations_manager	Initiates purchase from supplier
Receive Goods	wh_supervisor · wh_operator	Logs received quantities against PO
Generate Putaway	wh_supervisor	Creates tasks from GRN and assigns to operator
Execute Putaway	wh_operator	Moves goods to bin locations via mobile
AP Invoice / Payment	finance_officer	Records payable and payment

Flow 2 — Order-to-Cash (Sales Fulfillment)

The outbound order pipeline from sales order creation through picking, quality check, dispatch, and payment collection.

Outbound Fulfillment Workflow — WMS + TMS + FINANCE

Step 1 — Create Sales Order *(WMS)*

Operations Manager creates an SO specifying customer, delivery date, and ordered products. Status: `pending` .

Step 2 — Generate Pick Tasks *(WMS · INVENTORY)*

Supervisor generates pick tasks from the SO. Each task maps a product to its bin location. Tasks are assigned to Warehouse Operators.

Step 3 — Execute Picking *(WMS · INVENTORY)*

Operator picks items from bins and confirms each line. Reserved stock is deducted from INVENTORY. SO status moves to `packed` .

Step 4 — Quality Check *(WMS)*

Checker reviews all lines of the packed order against expected quantities.

- **FAIL** — returns to picking
- **PASS** — ready to dispatch

****v1.1 clarification — Checker permissions:**** The ``checker`` role has read access to both ``GET /wms/check-tasks`` and ``GET /wms/sales-orders/:id``. This is required for checkers to view the full SO details during QC. Prior to v1.1 the checker role lacked the ``wms:so:read`` permission and could not view order details.

Step 5 — Dispatch & Invoice *(WMS → TMS · FINANCE)*

Supervisor dispatches the SO — this triggers a TMS delivery route. WMS generates a Sales Invoice, which appears in Finance as an AR receivable.

Step 6 — Collect Payment *(FINANCE)*

Finance Officer records customer payment against the AR invoice. Aging report updated.

Roles Involved

Step	Role	Action
Create SO	operations_manager	Initiates customer order
Generate Picks	wh_supervisor	Auto-generates tasks per SO line
Execute Picks	wh_operator	Physical picking from bins
Quality Check	checker	Verify packed quantities and SO details
Dispatch	wh_supervisor · dispatcher	Release order to logistics
AR Payment	finance_officer	Records customer payment

Flow 3 — Last-Mile Delivery

The physical delivery execution from route creation through GPS tracking, proof of delivery, failed delivery resolution, and fuel logging.

Delivery Execution Workflow — TMS

Step 1 — Create Delivery Route *(TMS)*

Dispatcher creates a route with a vehicle, driver, and ordered delivery stops. Each stop is linked to a dispatched Sales Order.

Step 2 — Driver Starts Route *(TMS)*

Driver opens the mobile app, views today's route, and taps "Start Route." Route status changes to `in_progress`.

Step 3 — Real-Time GPS Tracking *(TMS)*

Mobile app sends GPS pings while driving. Operations team can monitor all active vehicles on the live map dashboard.

****Push notifications:**** When a new route is assigned to a driver, the system sends a push notification via Expo Push API (if the driver's device token is registered). Drivers register their token by calling `POST /tms/push-token` from the mobile app on login.

Step 4 — Confirm Delivery (POD) *(TMS)*

Driver confirms each stop. Two outcomes are possible:

4a — Delivered (success path)

Driver taps "Mark as Delivered" with optional notes and photo (Proof of Delivery). Stop status: `delivered`.

4b — Failed Delivery (v1.1 addition)

Driver taps "Mark as Failed." A reason field appears (required). On confirmation:

- Stop status changes to `failed`, `failure_reason` is recorded.
- A record is created in `failed_delivery_logs`.
- The linked Sales Order status changes to `delivery_failed`.

- The dispatcher sees an unresolved badge count on the Failed Deliveries section.
- Route completion checks only count `pending` stops — failed stops do not block route completion.

Step 5 — Failed Delivery Resolution *(TMS — v1.1 addition)*

Dispatcher views all unresolved failed stops in the Failed Deliveries panel and chooses one of two actions:

Action	What Happens
Reschedule	Dispatcher selects a target route (planned or active) and a stop sequence. A new <code>`pending`</code> stop is created on that route. The SO status returns to <code>`dispatched`</code> . The original failed stop is marked <code>`resolution: rescheduled`</code> .
Cancel	Dispatcher provides a cancellation reason. The SO is set to <code>`cancelled`</code> . Any outstanding AR invoice for the SO is voided. The failed stop is marked <code>`resolution: cancelled`</code> .

Once resolved, the stop disappears from the unresolved queue.

Step 6 — Log Fuel *(TMS)*

After completing the route, driver selects the completed route, enters liters used and distance driven. System computes fuel efficiency (km/L) automatically.

Roles Involved

Step	Role	Action
Create Route	dispatcher · operations_manager	Plan stops and assign driver/vehicle
Start Route	driver	Begin delivery on mobile app
GPS Tracking	driver (sends) · dispatcher (views)	Real-time vehicle monitoring
Confirm Delivery	driver	POD with photo per stop
Failed Delivery	driver	Record failure reason on mobile
Reschedule / Cancel	dispatcher · operations_manager	Resolve failed stops in web dashboard
Fuel Log	driver	Post-route fuel and distance entry

Flow 4 — Workforce Management

Employee lifecycle from onboarding through daily shift scheduling, attendance clock-in/out with late detection, and login audit visibility.

HRIS Workflow — HRIS

Step 1 — Create Employee Account *(HRIS)*

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HR Manager creates a user record with employee code, full name, department, and assigned role. The role controls what the employee can access across all modules.

Step 2 — Assign Shifts *(HRIS)*

HR creates shift records (start time, end time, shift type) for employees. Shift types: `regular` , `overtime` , `rest_day` , `holiday` .

Step 3 — Daily Attendance & Late Detection *(HRIS — v1.1 addition)*

Employees clock in and out via `POST /hris/attendance` . The system computes attendance status automatically:

Scenario	Status Assigned
Clock-in within 15 minutes of scheduled start	<code>`present`</code>
Clock-in more than 15 minutes after scheduled start	<code>`late`</code> (with <code>`late_minutes`</code> recorded)
Shift start has passed by > 15 min, no clock-in recorded	<code>`absent`</code> (set by mark-absent job)

Late detection logic:

- On clock-in, the system looks up the employee's shift for `CURRENT_DATE` .
- It computes `diff_minutes = NOW() - (shift_date + start_time)` .
- If `diff_minutes > 15` , shift status is updated to `late` and `late_minutes` is stored.
- The `clock_in` and `clock_out` timestamps are stored on the shift record (not just in `attendance_logs`).

Absent marking:

- HR or Operations Manager triggers `POST /hris/attendance/mark-absent` .
- The system updates all shifts for today where `clock_in IS NULL` and `start_time + 15 min` has already passed to `status = absent` .
- This is a manual trigger (intended to be run end-of-business-day).

Step 4 — Login Audit *(HRIS)*

Every login is recorded (timestamp, device type, IP address). HR and Operations Managers can view login history per employee with device icons for quick identification.

Role Hierarchy & System-Wide Access

Role	Modules	Key Responsibilities
<code>system_admin</code>	All modules	Full access to every operation and record
<code>operations_manager</code>	HRIS, Inventory, WMS, TMS, Finance	Create orders, manage inventory, oversee all operations
<code>hr_manager</code> · <code>hr_staff</code>	HRIS	Employee records, shift scheduling, attendance
<code>wh_supervisor</code>	Inventory, WMS	Generate pick/putaway/check tasks, dispatch orders

wh_operator	WMS, Inventory	Execute picks, receive goods, putaway
checker	WMS	Quality control of packed orders — read access to check tasks and SO details
dispatcher	WMS, TMS	Create delivery routes, monitor GPS, manage vehicles, resolve failed deliveries
driver	TMS	Start routes, confirm deliveries (POD or failed), log fuel
finance_officer	Finance	AP/AR invoice management and payment recording

Appendix — Key Status Lifecycles

Sales Order Status

```
pending → packed → dispatched → delivered
                                     → delivery_failed → dispatched (rescheduled)
                                     → cancelled
```

Putaway Task Status

```
pending → completed
```

Delivery Stop Status

```
pending → delivered
        → failed → (resolution: rescheduled | cancelled)
```

Shift / Attendance Status

```
present (default) → present  (clocked in on time)
                   → late    (clocked in past grace period)
                   → absent   (never clocked in, end-of-day job)
```

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